Day 3: Peripheral Troubleshooting

Learning Objectives

After this session, students will be able to:

- Identify common peripheral problems.
- Understand troubleshooting steps.
- Fix printer, scanner, keyboard, mouse, and webcam issues.
- Apply logical troubleshooting methods in real-life situations.

Theory Session

What is Peripheral Troubleshooting?

Peripheral troubleshooting means finding and fixing problems in external devices connected to a computer.

Examples:

- Printer not printing
- Webcam not detected
- Keyboard not typing
- Mouse not working
- Scanner not scanning

Basic Troubleshooting Flow

```
Identify Problem
        ↓
Check Connections
        ↓
```

```
Check Power Supply
        ↓
Check Drivers
        ↓
Restart Device
        ↓
Test Device
        ↓
Replace Hardware if Needed
```

Common Peripheral Issues and Solutions

1. Printer Not Printing

Possible Causes

- Printer is OFF
- USB cable disconnected
- No paper
- Paper jam
- Driver missing
- Printer Offline

Solutions

✔ Check power cable

✔ Check USB/Wi-Fi connection

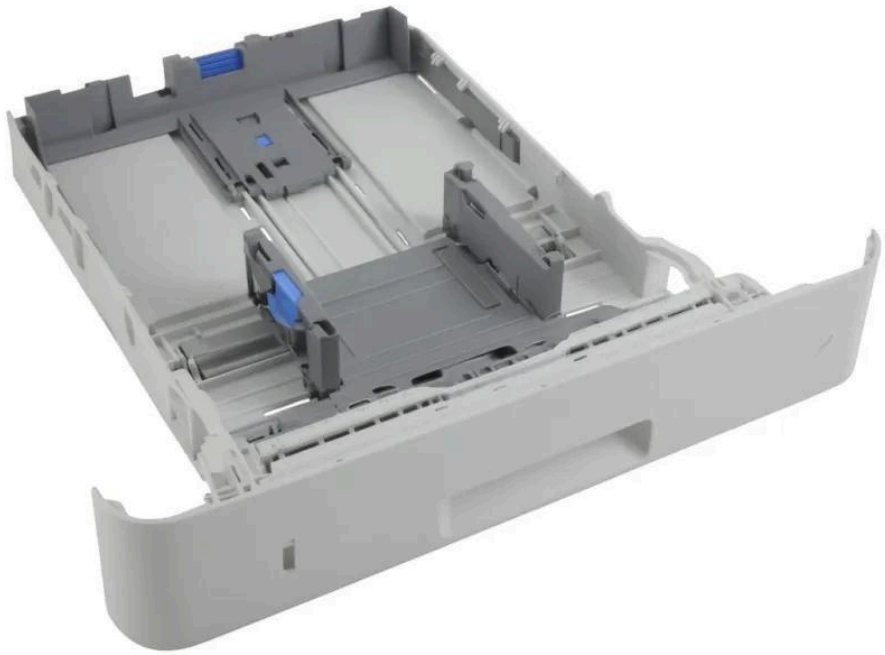
✔ Load paper

✔ Remove paper jam

✔ Restart printer

✔ Reinstall printer driver

Printer Components






ID Copy
Reduce Enlarge
Paper Select
Darkness
Ready To Copy
Tray2
OK
Menu
Back
Status
Stop/Clear
Power/ Wakeup
Start
Reset



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2. Keyboard Not Working

Causes

- Loose USB cable
- Driver issue
- Dust inside keyboard

Solutions

✔ Reconnect keyboard

✔ Try another USB port

✔ Restart computer

✔ Update drivers

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3. Mouse Not Working

Causes

- USB problem
- Battery dead (wireless mouse)
- Surface issue

Solutions

✔ Change batteries

✔ Use mouse pad

✔ Try another USB port

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4. Webcam Not Working

Causes

- Driver missing
- Camera blocked by privacy settings
- USB issue

Solutions

✔ Check camera permissions

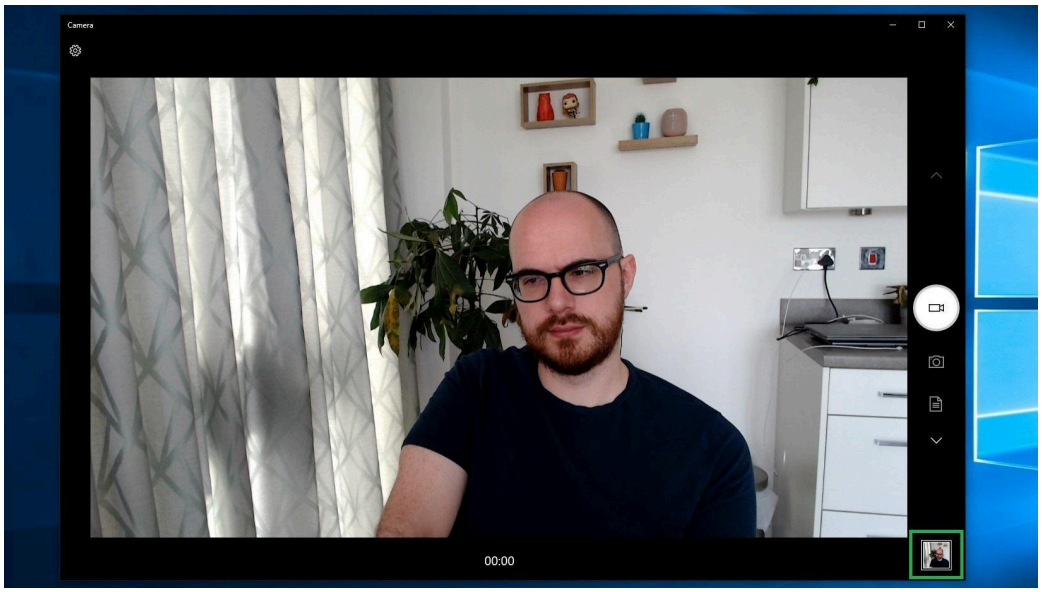
✔ Reconnect webcam

✔ Update driver

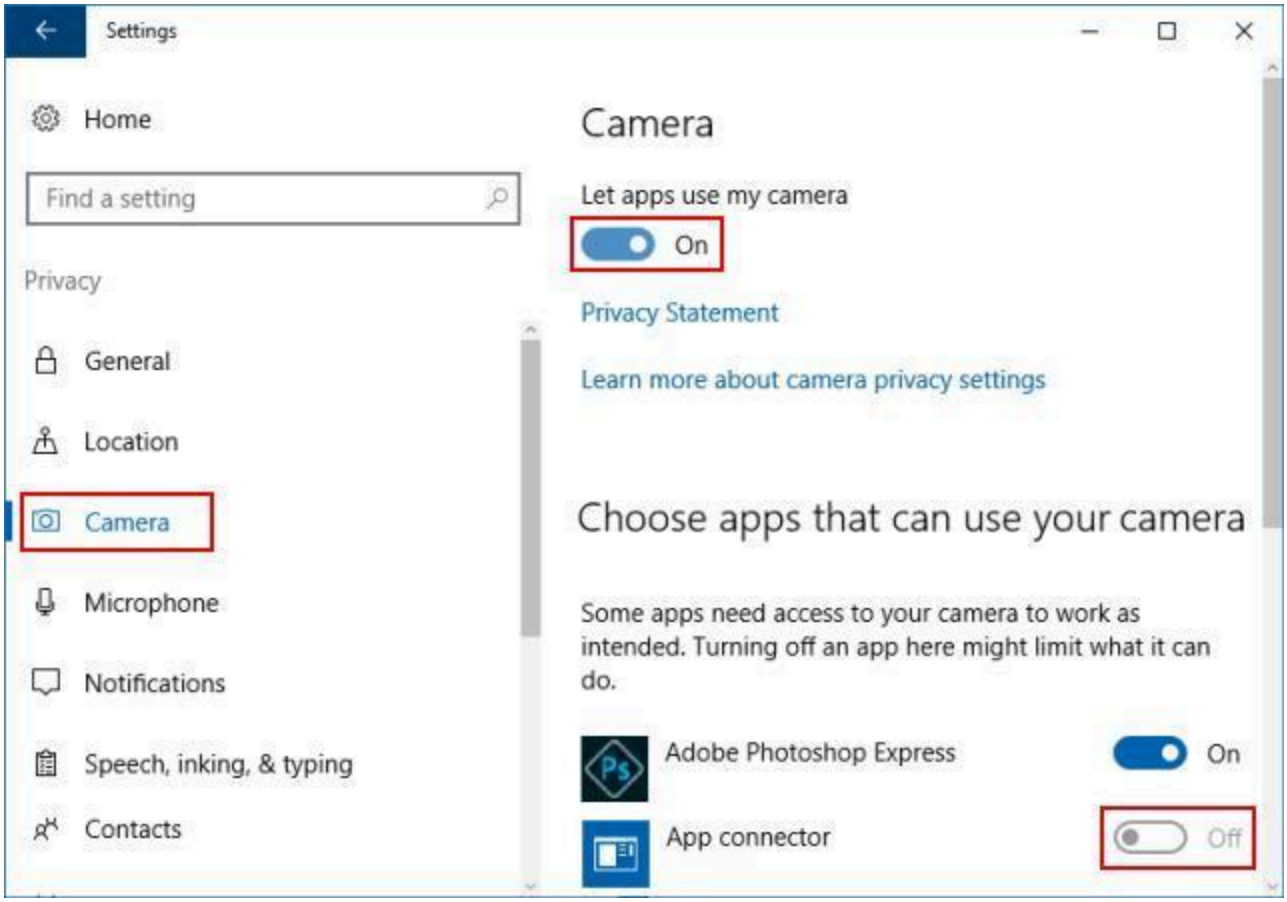
✔ Test in Camera Application

Webcam Examples

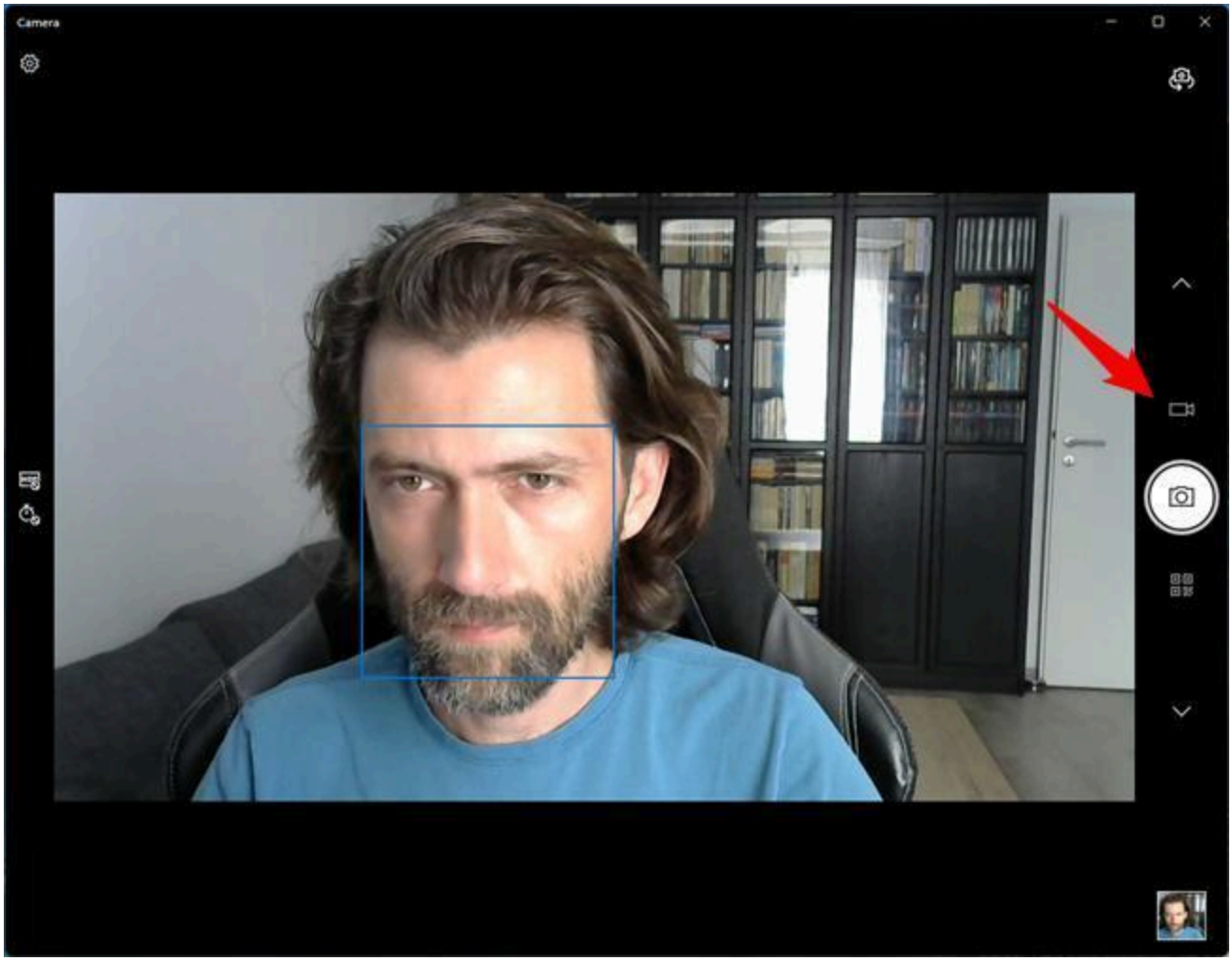





1: Open the base of the camera and place it on the screen
2: Adjust the base to securely clip the camera onto the screen
3:Connect the USB cable to the webcam
4: Plug the USB into the device


Settings
Home
Find a setting
Privacy
General
Location
Camera
Microphone
Notifications
Speech, inking, & typing
Contacts
Camera
Let apps use my camera
On
Privacy Statement
Learn more about camera privacy settings
Choose apps that can use your camera
Some apps need access to your camera to work as intended. Turning off an app here might limit what it can do.
Adobe Photoshop Express
On
App connector
Off


External Webcam?
Plug into 1st USB Port
External Microphone?
Plug into 2nd USB Port



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5. Scanner Not Working

Causes

- Driver missing
- USB disconnected
- Scanner software problem

Solutions

✔ Check cable

✔ Restart scanner

✔ Reinstall software

✔ Test with Windows Scan App

Real-Life Troubleshooting Tips

Rule 1

Always check physical connections first.

Rule 2

Restart device before making major changes.

Rule 3

Check Device Manager for errors.

Rule 4

Test device on another computer.

Rule 5

Never assume hardware is faulty immediately.

Practical Session

Practical 1: Troubleshoot Non-Working Printer

Scenario

Student reports:

"Printer is connected but not printing."

Step 1

Press:

```
Win + R
```

Type:

```
control printers
```

Press Enter.

Step 2

Check printer status.

Possible status:

```
Offline
Paused
Ready
Error
```

Step 3

Right-click Printer

Select:

```
See What's Printing
```

Remove stuck jobs:

```
Cancel All Documents
```

Step 4

Check USB Cable

Disconnect and reconnect.

Step 5

Restart Printer

Turn OFF

Wait 10 seconds

Turn ON

Step 6

Print Test Page

```
Printer Properties
→ Print Test Page
```

Step 7

Check Device Manager

```
Win + X
→ Device Manager
```

Look for:

```
Yellow Warning Symbol
```

Expected Result

Printer should successfully print test page.

Day 4: Connecting and Testing Peripherals

Learning Objectives

Students will:

- Connect peripherals properly.
- Install required drivers.
- Test devices.
- Solve connectivity issues.

Theory

Steps to Connect Any Peripheral

Step 1

Connect device physically.

Step 2

Power ON device.

Step 3

Install drivers if required.

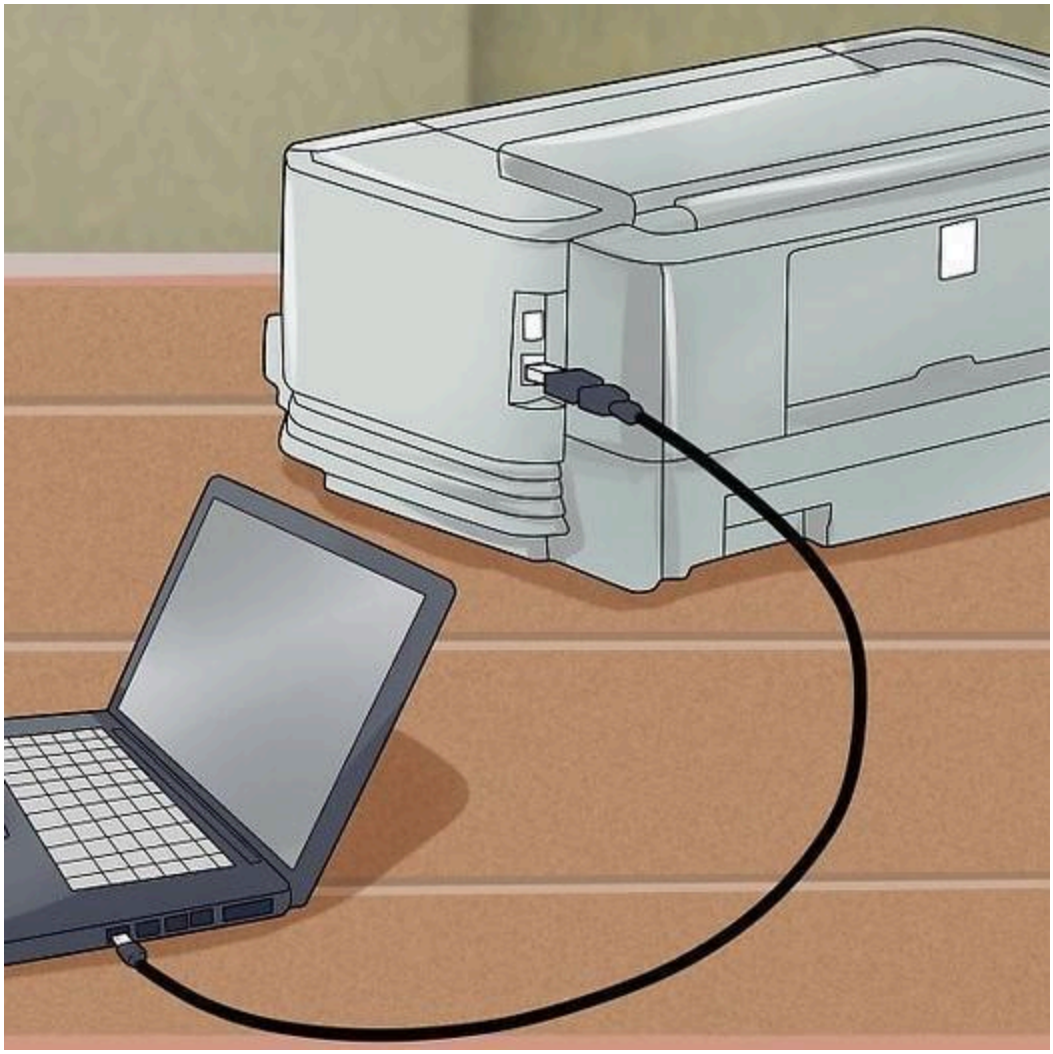
Step 4

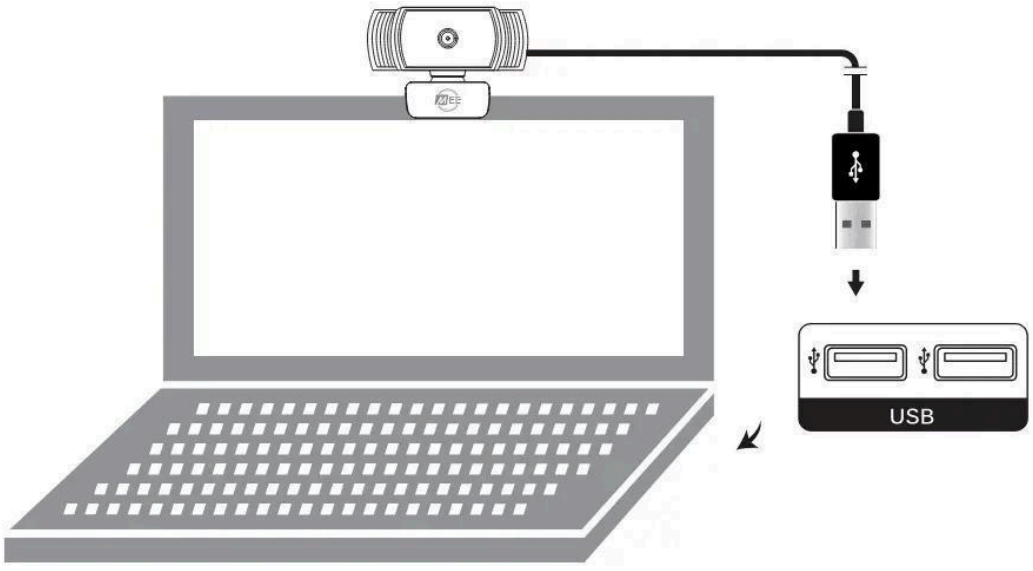
Verify detection in Windows.

Step 5

Test functionality.

Common Peripheral Connections



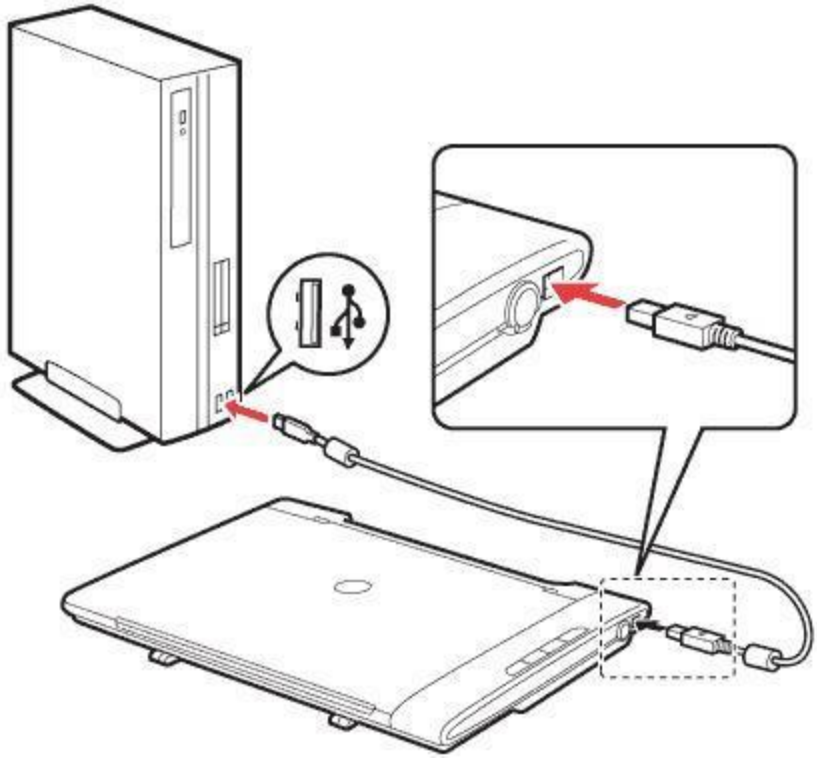

USB


USB






1: Open the base of the camera and place it on the screen
2: Adjust the base to securely clip the camera onto the screen
3:Connect the USB cable to the webcam
4: Plug the USB into the device



Practical Session

Practical 1: Setup and Test Printer

Tasks

1. Connect printer USB cable.
2. Power ON printer.
3. Install printer driver.
4. Open:

```
Control Panel
→ Devices and Printers
```

5. Print Test Page.

Success Criteria

Printer prints test page successfully.

Practical 2: Setup and Test Webcam

Tasks

Connect webcam.

Open:

`Camera App`

Verify live video.

Success Criteria

Video feed visible.

Practical 3: Setup and Test Scanner

Tasks

Connect scanner.

Open:

`Windows Scan`

Scan one document.

Save as PDF.

Success Criteria

Document scanned successfully.

Real-Time Lab Tasks (Interesting Activities)

Task 1: Mystery Printer Problem

Instructor secretly:

- Disconnect printer USB cable

OR

- Set printer Offline

Students must find and fix issue.

Task 2: Webcam Detective

Instructor:

- Disable webcam from Device Manager

Students must:

- Find issue
- Enable device
- Test camera

Task 3: Driver Missing Challenge

Uninstall a webcam/printer driver.

Students must:

- Open Device Manager
- Identify unknown device
- Reinstall driver

Task 4: Wrong USB Port Challenge

Connect peripheral to faulty USB port.

Students must:

- Diagnose issue
- Test another port
- Verify working condition

Task 5: IT Support Engineer Simulation

Scenario:

Client says printer is not printing and webcam is not detected.

Students must create a troubleshooting report:

Check	Status
Power Supply	OK
Cable Connection	OK
Driver Installed	OK
Device Detected	OK
Test Successful	YES

Bonus Practical (Students Love This)

View All Connected Devices

Open CMD:

```
driverquery
```

Shows installed drivers.

Check USB Devices

Open PowerShell:

```
Get-PnpDevice
```

Shows all connected hardware.

Open Device Manager Quickly

```
devmgmt.msc
```

Print Queue Shortcut

```
control printers
```

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1. Matrix Screen Trick 😎

Open CMD and type:

```
color 0A
tree C:\
```

Or save this in Notepad:

```
@echo off
color 0A
:a
echo %random%%random%%random%%random%
goto a
```

Save as:

```
matrix.bat
```

Students will see a screen similar to the one in the *Matrix* movie.

---

2. Fake Virus Prank

Write this in Notepad:

```
@echo off
msg * Virus Detected!
msg * System Scan Started!
msg * Just Kidding :)
```

Save as:

```
prank.bat
```

Note: For educational purposes only.

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3. Personal Computer Information

In CMD:

```
systeminfo
```

Challenge:

- How much RAM does the computer have?
- What is the installation date?
- Which Windows version is installed?

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4. Wi-Fi Password Reveal

Open CMD as Administrator:

```
netsh wlan show profiles
```

Then:

```
netsh wlan show profile name="WiFi_Name"
key=clear
```

Find:

```
Key Content
```

Learning: Understand where stored Wi-Fi passwords are displayed.

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5. Hidden God Mode

Create a new folder on the Desktop:

```
GodMode.{ED7BA470-8E54-465E-825C-99712043E01C
}
```

An instant Windows Control Center will appear.

Students will find it magical 😄

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6. Secret Clipboard History

Press:

`Win + V`

Clipboard History will open.

Challenge: Copy 5 different pieces of text and test it.

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7. Emoji Panel

Press:

`Win + .`

The Windows Emoji Panel will open.

Useful for LinkedIn, Teams, emails, and more.

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8. Record Your Screen Without Software

Press:

`Win + G`

Xbox Game Bar will open.

Challenge: Record your screen for 10 seconds.

---

9. Shutdown Timer

In CMD:

```
shutdown -s -t 60
```

To cancel:

```
shutdown -a
```

Learning: Introduction to system administration concepts.

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10. Open Websites From CMD

In CMD:

```
start google.com
```

Or:

```
start youtube.com
```

Students may find this interesting.

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11. Create a Folder With a Reserved Name

In CMD:

```
mkdir \\?\C:\con
```

Explain the concept of reserved Windows device names:

- `CON`
- `PRN`
- `AUX`
- `NUL`

---

12. Generate a Battery Report

On a laptop:

```
powercfg /batteryreport
```

A battery report will be generated.

Challenge: Find the battery health information.

---

13. See Who Is Logged In

In CMD:

```
query user
```

Or:

```
whoami
```

Learning: Understanding user accounts and logged-in users.

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14. Find Open Network Connections

In CMD:

```
netstat -ano
```

Challenge: How many active connections are there?

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15. Open the Secret Character Map

Run:

```
charmap
```

Copy and explore special symbols.

Useful for learning about character sets and symbols.

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16. Keyboard Shortcut Race

Shortcut	Action
Win + E	File Explorer
Win + R	Run
Win + D	Show Desktop
Win + L	Lock Computer
Alt + Tab	Switch Apps
Ctrl + Shift + Esc	Task Manager

Challenge: The fastest student wins.

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17. Create Your Own Command

Open Notepad and write:

```
@echo off
echo Welcome to ABC Institute
pause
```

Save as:

```
welcome.bat
```

Double-click to run it.

---

18. Network Detective

In CMD:

```
ipconfig
ping google.com
tracert google.com
```

Challenge: Find out how many hops the connection takes to reach Google.

---

20. RAM Consumption Challenge

Open Task Manager:

```
Ctrl + Shift + Esc
```

Challenge:

- Which application is using the most RAM?
- Which application is using the most CPU?